

Zain Sadiq

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SUMMARY

Geography and data focused undergraduate with experience editing, processing, and analyzing geospatial datasets using various tools like ArcGIS Pro, and ERDAS IMAGINE. Skilled in Land Surface Temperature derivation, multi-source spatial analysis, cartography, and vulnerability index construction. Experienced in translating satellite imagery and demographic data into publication-quality maps and technical reports, and data visualizations with an emphasis on environmental justice applications.

TECHNICAL SKILLS

• ArcGIS, Spatial Analysis, Cartography, Map Design, GIS Data Visualization, Excel, Remote Sensing, Tableau, R Studio, Fluent in Japanese, ERDAS IMAGINE, GIS Data Editing & Maintenance, Python, MS Office

PROJECTS

Urban Heat Island & Vulnerability Analysis - Phoenix, AZ

- Derived Land Surface Temperature and NDVI from Landsat 9 Collection 2 Level-2 imagery for Maricopa County, AZ using radiometric scale factor conversion and multispectral band math
- Constructed a composite Heat Vulnerability Index via min-max normalization and weighted overlay of six physical and socioeconomic indicators including LST, impervious surface density, median income, age, and race
- Produced five publication quality cartographic outputs and a full technical report framing findings within an environmental justice context

Flood Risk + Social Vulnerability - Miami-Dade Florida

- Edited multiple geospatial datasets of FEMA flood hazard layers, census tract boundaries, and social vulnerability to build a compound risk model
- Resolved spatial conflicts and data inconsistencies across source layers to ensure accurate feature alignment and attribute integrity
- Produced a bivariate choropleth map identifying 110 high compound-risk tracts, supporting equitable disaster resilience planning

NYC NDVI Vegetation Analysis

- Processed and modified multi-year Landsat imagery in ERDAS IMAGINE to extract vegetation data from 2010 to 2024
- Edited and reclassified raster datasets to isolate land cover change, correcting for atmospheric and sensor inconsistencies

Mars Landing Sites Visualization

- Edited and classified planetary datasets using ArcGIS Pro symbology and spatial classification tools
- Modified attribute tables and feature layers to support environmental interpretation and site suitability outputs

NYC Housing Prices Visualization

- Compiled and cleaned housing price data across NYC's five boroughs for use in an interactive Tableau dashboard
- Produced charts, statistics, and map-based visualizations for comparative spatial analysis

EXPERIENCE

Rutgers University - Aresty

May 2026 - Present

GIS Undergraduate Research Assistant

- Editing, processing, and analyzing geospatial datasets to support ongoing volcano morphology research under faculty direction
- Modifying and maintaining spatial data layers in ArcGIS Pro and ERDAS IMAGINE to ensure accuracy across multi-source datasets
- Performing spatial analysis and producing visualizations in Tableau and Excel to communicate research findings

The Commencement Group

May 2026 - Jun 2026

Outside Sales Intern

- Conducted direct, face-to-face client outreach as part of a field-based sales team
- Tracked and maintained inventory records using Excel and Google Sheets

EDUCATION

Rutgers University

Sep 2023 - May 2027

B.A., Information Technology Informatics | Data Track

New Brunswick

- GPA: 3.841

Rutgers University

Sep 2023 - May 2027

B.A., Geography | Tech Track

New Brunswick

- GPA: 3.417

Rutgers University

Sep 2023 - May 2027

Minor, Data Science

New Brunswick

- GPA: 3.61